
CLEW

Classification with Learned Embeddings to Compute the Wasserstein Distance

A Model-Agnostic Metric for
Generative Image Evaluation

Felix Blanchard¹

Anh-Khoa Ngo-Ho²

¹ Institut Polytechnique de Paris

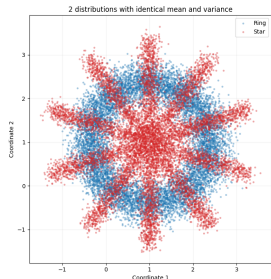
² Nam Can Tho University

Context

Why another generative metric?

- ▶ FID, KID, and generative precision/recall rely strongly on **Euclidean geometry** in an embedding space.
- ▶ Standard Inception features are optimized for **ImageNet classification**, not necessarily for semantic realism.
- ▶ Two distributions can have similar global statistics while having very different underlying structures.
- ▶ Moving to new datasets is difficult when **labels are unavailable or expensive**.
- ▶ Most existing metrics measure distances between distributions, they do not provide **per-sample information**.

Limitations of existing metrics



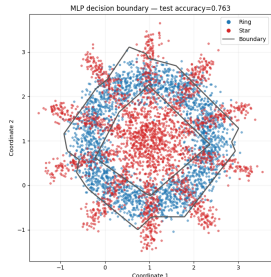
Example of two clearly distinguishable datasets despite near-perfect scores under Fréchet Distance and Maximum Mean Discrepancy, and a Generative Precision ($k = 5$) of 0.935.

Proposed solution

Main elements of the solution

- ▶ Leverage pretrained **self-supervised encoders** (I-JEPA, DINOv2, CLIP) to produce semantically rich image embeddings.
- ▶ Train a small **classifier**, using a k -fold scheme so that every image is included in a held-out test set, to distinguish between real and generated images.
- ▶ Compute the **Wasserstein distance** between the classifier-output probability distributions for real and generated samples.
- ▶ Recover a bounded $[0, 1]$, easily interpretable score that can also provide a **per-sample result**.

Why a classifier?



A simple neural network (one hidden layer with 16 neurons) better captures the difference between the two manifolds than standard generative evaluation metrics.